

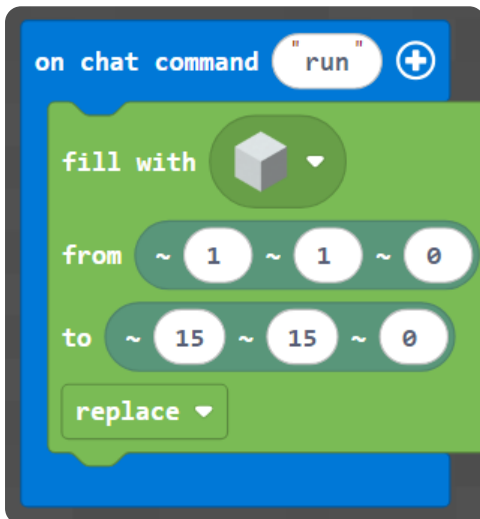
3D Coordinates Extension — English Worksheet (Advanced)

Topic: Design Your Own Pixel Art (x, y) · **Course:** 3D Coordinates · **Level:** Advanced (Extension, after Lesson 3) · **Time:** about 45 minutes

Every other week the picture was given to you. This lesson **you** draw it — then **you** read your own drawing back onto the wall.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your home spot:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too.

1 · Predict 🌟

Read each set of steps. Write what you will see **and the reason**.

```
place blue block at (3, 9)
place blue block at (4, 9)
place blue block at (5, 9)
```

Three blocks share $y = 9$. Row or column? Which number stays the same, and why?


```
place blue block at (7, 2)
place blue block at (7, 3)
place blue block at (7, 4)
place blue block at (7, 5)
```

These four share $x = 7$. Row or column? Which number changes, and why?


```
place green block at (5, 6)
place green block at (6, 6)
place green block at (5, 7)
place green block at (6, 7)
```

These four change both numbers. What shape do they make, and why?


```
place green block at (6, 6)
place blue block at (6, 6)
```

Two blocks aim for the same (x, y) . What ends up on the wall, and why?

2 · Spot the Bug 🐛

Each one is broken. Fix it, then explain why the original was wrong and why your fix works.

Bug A — The flower's two leaves should sit at (2, 4) and (4, 4). Both land in the **same spot**.

```
place green block at (2, 4)
place green block at (2, 4)
```

Write the fixed line:

Why was it wrong? Why does your fix work?

Bug B — This should be a column going **up** at $x = 8$, from $y = 5$ to $y = 7$.

```
place black block at (8, 5)
place black block at (9, 5)
place black block at (10, 5)
```

Write the fixed lines:

Why was it wrong? Why does your fix work?

Bug C — This should be a 2×2 square with its bottom-left corner at (4, 4).

```
place blue block at (4, 4)
place blue block at (5, 4)
place blue block at (4, 4)
place blue block at (5, 4)
```

Write the fixed lines:

Why was it wrong? Why does your fix work?

Stand on your **home spot** (the red block).

Your five rules.

#	Your design must...
1	fit inside the 15 × 15 wall
2	use 3 or more different block colours
3	use 25 or more blocks
4	be recognisable — someone else can name it without being told
5	be drawn on the grid below before you place a single block

Draw your design. Colour the squares. x goes **across**, y goes **up**.

 Your design

Write the plan. List the (x, y) of every block in your **biggest** colour, in the order you will place them.

Then build it. Read each coloured square's (x, y) and place that block on your wall.

Check your work. Hold your grid next to your wall. Write down every block you had to move, and what was wrong with your first (x, y).

Someone else names it. Show your build to a person who has not seen your grid. What did they say it was? If they got it wrong, what would you change?

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your grid. Hold this paper up to the camera.

2 • Your build. Hold your paper next to the wall so both are in the shot.

3 • Your five rules. Check each one out loud against your build.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.